

Use-Case Flow Specification

Problem Statement #24: Public Bus Live Tracking & Crowding Estimator

Use Case: UC-02 Search Stop & View ETA

Primary Actor: Commuter

Includes: UC-05 Calculate ETA

Preconditions:

- The commuter has the app open and location services / stop search is available.
- The target stop exists in the system's stop database and has at least one route serving it.
- The system is receiving live GPS telemetry from buses on the relevant route(s).

Postconditions:

- The commuter is shown an up-to-date list of approaching buses for the selected stop, each with a live ETA and crowding level.
- The displayed ETA reflects the most recent GPS-based recalculation (UC-05).

Main Success Scenario:

1. Commuter opens the app and selects "Find my stop."
2. System prompts the commuter to search by stop name/number or to use current location.
3. Commuter selects a valid stop.
4. System retrieves all routes and active buses currently approaching the selected stop.
5. System invokes UC-05 (Calculate ETA) for each approaching bus, using its latest GPS position, route geometry, and current traffic speed.
6. System retrieves the latest crowding level for each bus (from FR-002).
7. System displays a list of approaching buses sorted by ETA, each showing route number, ETA in minutes, and crowding level (Low/Medium/High).
8. System refreshes the ETA list using the latest available GPS data while the commuter remains on the screen.
9. Commuter reviews the list and selects a bus to track or favourite (optional, leads to UC-01/UC-06).
10. Use case ends successfully.

Alternate Flow — 4a. No Buses Currently Approaching the Stop:

- 4a1. System finds no active bus within the configured ETA window for the selected stop.
- 4a2. System displays a "No buses currently approaching this stop" message along with the next scheduled departure time, if known.
- 4a3. Commuter may choose to select a different stop, returning to step 2 of the main flow, or exit the use case.

Alternate Flow — 5a. GPS Signal Lost for an Approaching Bus:

- 5a1. System detects that a bus has not sent a GPS update for more than 30 seconds.
- 5a2. System marks that bus's ETA as "Estimated (signal delayed)" using its last known position and speed instead of removing it from the list.
- 5a3. If signal is not restored within 2 minutes, the bus is removed from the live list and the commuter is shown a brief notice that tracking was temporarily interrupted.